MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3ADP1, 3ADP2, 3ADP7, AND 3ADP11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, flat face with post accelerator

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio	ft
Maximum:	6.9	0	1,100	3,300	6,600	600	1.5	1.0 (Note 1)	±180	2.0 (Note 2)	50,000
Minimum:	5.7	-200	---	1,000	1,000	---	---	---	---	---	---
TEST CONDITIONS: (See note 3)	6.3	Adjust	Focus	2,000	4,000	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
(E)	Qualification inspection				---	---	---	---
1141	Pressure (implosion)	All	-		---	---	---	---
1002	Barometric pressure, reduced	All	5	87 ±4 mmHg				
5111	Vibration	All	-		Width	---	1.0	mm
1331	Direct-interelectrode capacitance	All	-	k to all	Ck	---	4.5	pF
				g1 to all	Cg1	---	7.2	pF
				D1 to D2	C1D2	---	2.0	pF
				D3 to D4	C3D4	---	1.6	pF
				D1 to all	CD1	---	4.6	pF
				D2 to all	CD2	---	4.6	pF
				D3 to all	CD3	---	3.6	pF
				D4 to all	CD4	---	3.6	pF
5101	Neck and bulb alinement (electrostatic types)	All	-		Dia	---	1.87	inch
5216	Cathode illumination	All	-		---	---	---	---
5246	Deflection factor	All	-	1D2; Eb3 = Eb2 = 2,000 Vdc	DF	108	132	Vdc/in.
5248	Deflection factor	All	-	3D4; Eb3 = Eb2 = 2,000 Vdc	DF	49	60	Vdc/in.
5248	Deflection-factor uniformity	All	-	1D2	---	---	---	---
1216	Base material insulating quality	All	-		---	---	---	---
	Quality conformance inspection, part 1							
5221	Light output	P1	8	Ib3 = 25 µAdc	Light	25	---	ftL
		P11	8	Ib3 = 25 µAdc	Light	14	---	ftL
5201	Voltage breakdown	All	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	All	-		---	---	---	---
5206	Gas "cross"	All	4	Ib3 = 25 Adc	---	---	---	---
5106	Screen and faceplate blemishes	All	-		---	---	---	---
5223	Modulation	All	-	Ib3 = 25 µAdc	ΔEc1	---	38	Vdc
5231	Spot position (electrostatic deflection)	All	-		---	---	10	mm
5231	Spot displacement (leakage)	All	-		Displ	---	8	mm
5241	Gr utoff voltage	All	-		Ec1	-87	-52	Vdc
---	Minimum useful scan	All	6		---	---	---	---
5103	Pattern distortion	All	7		---	---	2.0	%
5251	Grid No. 1 leakage current	All	-		---	---	---	---
5251	Anode No. 1 leakage current	All	-		---	---	---	---
5251	Anode No. 2 leakage current	All	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2							
5201	Electrode current (cathode)	All	-	Ib3 = 25 µAdc	Ik	---	500	µAdc
5201	Electrode current (anode No. 1)	All	-		Ib1	-15	+5	µAdc
1301	Heater current	All	-		If	540	660	mAdc
5101	Angle between traces	All	-		---	89	91	Degrees
5216	Stray light emission (conventional types)	All	-	Eb2 = 3,300 Vdc; Eb3 = 6,600 Vdc	---	---	---	---
5226	Line width "A" (electrostatic deflection)	All	-	Ib3 = 25 µAdc	Width	---	0.35	mm
5246	Focusing voltage at cutoff	All	-		Eb1	---	490	Vdc
5101	Base alinement (electrostatic types)	All	-	1D2; pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	-	1D2	---	---	---	---
5101	Side terminal and base alinement	All	-	Pin No. 5	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All	-		---	---	---	---
5221	Screens	P7	-		---	---	---	---
		P2	9		cb1	360	---	cb
5226	Line width "B" (electrostatic deflection)	All	10	Ib3 = 25 µAdc	Width	---	0.50	mm
5246	Focusing voltage at modulation condition	All	-		Eb1	320	---	Vdc
5248	Deflection factor (1D2)	All	-		DF	133	163	Vdc/in.
5248	Deflection factor (3D4)	All	-		DF	59	73	Vdc/in.
---	Trace distortion	All	11		---	---	0.020	inch
5251	Heater-cathode leakage current	All	-		Ihk	---	15	µAdc
1101	Secureness of base, cap, or insert	All	-		---	---	---	---
(E) 1111	Base pin solder depth	All	-		---	---	---	---
(E) 1105	Permanence of marking	All	-		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	All	-	Group C; Ib3 = 25 Adc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:							
5223	Modulation	All	-	Ib3 = 20 µAdc	ΔEc1	---	38	Vdc
5226	Line width "A"	All	-	Ib3 = 20 µAdc	Width	---	0.35	mm
5226	Line width "B"	All	-	Ib3 = 20 µAdc	Width	---	0.50	mm

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NOTES:

1. It is recommended that the deflection electrode circuit resistances be approximately equal.
2. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of EB3/Eb2 may result in changes in deflection uniformity and pattern distortion.
3. All tests are to be performed with the tube shielded.
4. This test to be performed at the conclusion of the holding period.
5. This test is to be made with a maximum rated voltage (Eg1 maximum negative voltage) applied to the base pins.
6. Useful scan.
   a. The useful scan in the 1D2 direction is ±1.3125 (33.34 mm) inches from the tube face center.
   b. The useful scan in the 3D4 direction is ±0.75 (19.05 mm) inch from the tube face center.
7. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 2.295 (58.29 mm) - by 1.530 (38.86 mm) - inch rectangle, will fall within the area bounded by the 2.295 (58.29 mm) - by 1.530 (38.86 mm) - inch rectangle and an inscribed 2.205 (56.01 mm) - by 1.470 (37.34 mm) - inch rectangle.
8. Use a raster size of 2.25 (57.15 mm) by 1.50 (38.10 mm) inches, and a type 3 photronic cell without eye correction, calibrated in foot candles of illumination from a light source having a color temperature of 2,700°K.
9. Use a raster size of 2.25 (57.15 mm) by 1.50 (38.10 mm) inches and a beam current of 26.1 µAdc.
10. The line width "B" shall be measured at ±0.75 (19.05 mm) inch from the tube face center.
11. The focused sides of a raster pattern, centered with respect to the tube face and whose mean dimensions are 2.25 (57.15 mm) by 1.50 (38.10 mm) inches, shall not deviate from straight lines passing through the corners by more than the indicated value.

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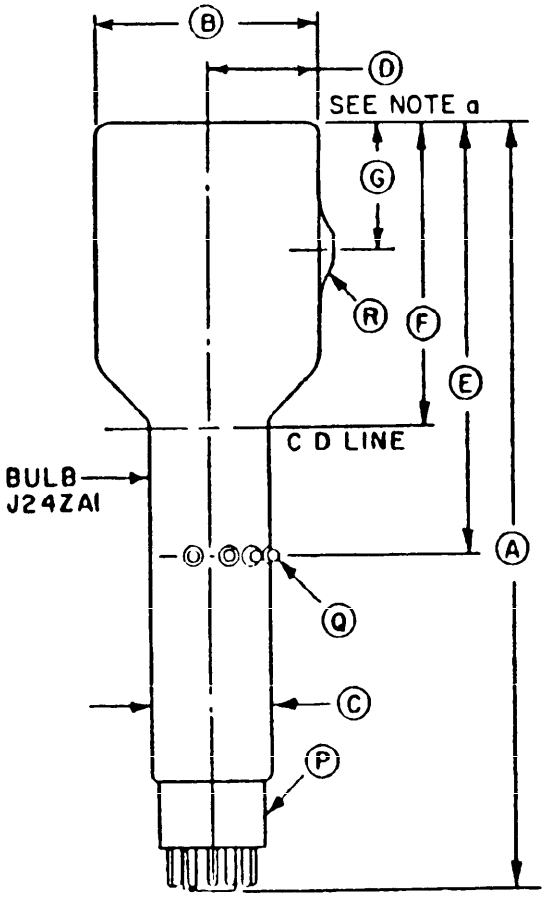
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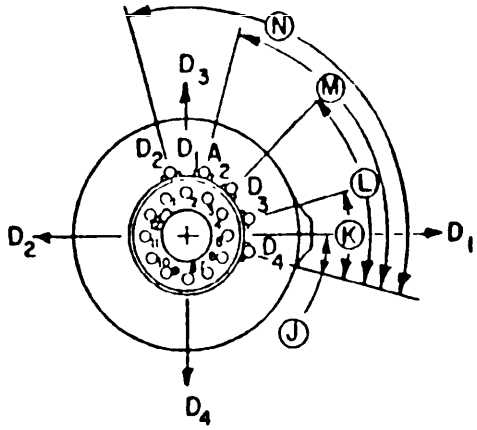
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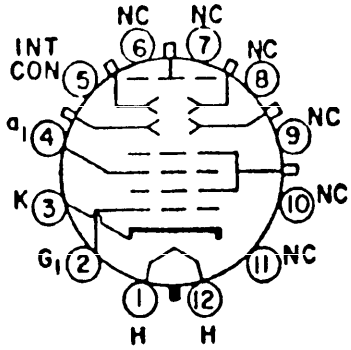
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BASE CONNECTIONS

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
QUALIFICATION INSPECTION		
P	Base: B12-43	
Q	Bulb contact: J1-25	
R	Bulb contact: J1-22	
Quality conformance inspection, part 2		
A	9.750 (247.65)	10.250 (260.35)
B	2.938 (74.63)	3.063 (77.80)
C	1.563 (39.70)	1.688 (42.88)
D	1.313 (33.35)	
E	5.188 (131.78)	5.563 (141.30)
F	3.688 (93.68)	4.063 (103.20)
G	1.313 (33.35)	1.688 (42.88)
J	10°	20°
K	25°	35°
L	55°	65°
M	85°	95°
N	115°	125°

Note:
a. Minimum useful screen radius.

FIGURE 1. Outline drawing of electron tube types 3ADP1, 3ADP2, 3ADP7, and 3ADP11.

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